

 B.TECH 1ST YEAR · 1ST SEMESTER

# BCME

Basic Civil & Mechanical Engineering

## ⚙️ UNIT II – Complete Revision Sheet

Manufacturing Processes + Thermal Engineering

 Manufacturing

 Casting

 Forming

 Joining

 Machining

 CNC

 3D Printing

 Boilers

 Thermodynamic Cycles

 Refrigeration

 IC Engines

 EV & Hybrid

✓ Exam-Oriented · ✓ Tables & Comparisons · ✓ Flashcards · ✓ Formulas



# 1. INTRODUCTION TO MANUFACTURING PROCESSES

## DEFINITION

**Manufacturing** = Process of converting raw materials into finished products using physical, chemical, or mechanical processes to add value.

## IMPORTANCE

- Converts raw materials → useful goods
- Creates employment & GDP growth
- Backbone of industrial economy
- Enables mass production & precision

## CLASSIFICATION

- **Primary Shaping** – casting, forging
- **Forming** – rolling, extrusion, drawing
- **Machining** – turning, drilling, milling
- **Joining** – welding, soldering, riveting
- **Surface Finishing** – grinding, painting

| PROCESS TYPE      | PURPOSE                            | EXAMPLES                             |
|-------------------|------------------------------------|--------------------------------------|
| Primary Shaping   | Give initial shape to raw material | Casting, Forging, Powder metallurgy  |
| Forming           | Deform material to desired shape   | Rolling, Extrusion, Drawing          |
| Machining         | Remove material for precision      | Turning, Drilling, Milling, Grinding |
| Joining           | Permanently/temporarily join parts | Welding, Brazing, Riveting           |
| Surface Finishing | Improve appearance/durability      | Plating, Painting, Anodizing         |



# 2. CASTING PROCESS

VERY IMPORTANT

## DEFINITION

**Casting** = Process of pouring molten metal into a mould of desired shape and allowing it to solidify to obtain the required shape.

## BASIC STEPS



## COMPONENTS

- **Pattern** – Replica of the part to be cast (wood/metal/plastic)
- **Mould** – Cavity into which molten metal is poured
- **Core** – Insert to create hollow sections in casting
- **Sprue** – Vertical channel for metal entry
- **Gate** – Passage connecting sprue to mould cavity
- **Riser** – Reservoir to compensate for shrinkage

## TYPES

- **Sand Casting** – Most common, low cost, any metal
- **Die Casting** – High pressure, permanent mould, high accuracy
- **Investment Casting** – Lost wax, complex shapes
- **Centrifugal Casting** – Rotating mould, pipes/cylinders

| FEATURE | ADVANTAGES | LIMITATIONS |
|---------|------------|-------------|
|---------|------------|-------------|

|            |                                   |                                   |
|------------|-----------------------------------|-----------------------------------|
| Cost       | Low tooling cost for sand casting | Die casting has high initial cost |
| Complexity | Complex shapes easily made        | Dimensional accuracy lower        |
| Size       | Very large parts possible         | Surface finish is rough           |
| Material   | Any metal can be cast             | Porosity & defects possible       |



3. FORMING PROCESSES

DEFINITION

**Forming** = Plastic deformation of material using mechanical force without removing material. Shape change through stress.

| PROCESS   | PRINCIPLE                                    | PRODUCT / APPLICATION                       |
|-----------|----------------------------------------------|---------------------------------------------|
| Forging   | Compressive force between dies; hammer blows | Crankshaft, connecting rod, gears, spanners |
| Rolling   | Material passed between rotating rolls       | Rails, plates, I-beams, sheets              |
| Extrusion | Material forced through a die opening        | Pipes, rods, channels, window frames        |
| Drawing   | Wire/rod pulled through a die                | Wires, tubes, cups (deep drawing)           |

 **HOT WORKING**

- Temperature: Above recrystallization temp
- No work hardening
- Less force required
- Poor surface finish
- Ex: Hot rolling, hot forging

 **COLD WORKING**

- Temperature: Below recrystallization temp
- Work hardening occurs
- More force required
- Good surface finish, better accuracy
- Ex: Cold drawing, cold rolling



4. JOINING PROCESSES

VERY IMPORTANT

DEFINITION

**Joining** = Process of permanently or temporarily connecting two or more parts together using heat, pressure, or adhesive materials.

| PROCESS   | TEMPERATURE         | FILLER MATERIAL     | STRENGTH  | USE                           |
|-----------|---------------------|---------------------|-----------|-------------------------------|
| Welding   | Very High (>1500°C) | Same as base metal  | Very High | Structural, ships, pipelines  |
| Brazing   | High (>450°C)       | Brass/bronze filler | High      | Thin parts, dissimilar metals |
| Soldering | Low (<450°C)        | Tin-lead alloy      | Low       | Electronics, PCB, wiring      |
| Riveting  | Room Temp           | Rivet (metal pin)   | Medium    | Aircraft, bridges, boilers    |

WELDING TYPES

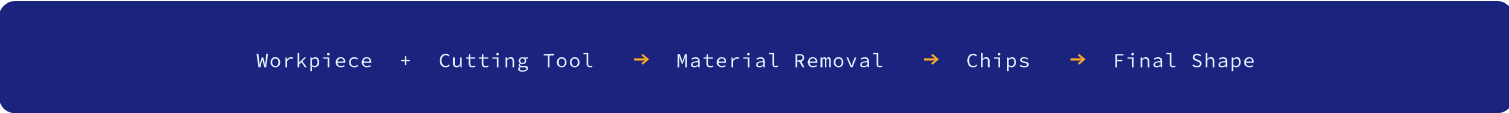
- **Arc Welding** – Electric arc (most common)
  - **MIG Welding** – Metal Inert Gas (fast, automated)
  - **TIG Welding** – Tungsten Inert Gas (precision)
- **Spot Welding** – Resistance heat (car bodies)
  - **Gas Welding** – Oxy-acetylene flame
  - **Laser Welding** – Precision, high-tech



5. MACHINING PROCESSES

DEFINITION

**Machining** = Material removal process using cutting tools to achieve desired shape, size, and surface finish. Subtractive manufacturing.



| Operation | Machine         | Movement                      | Application                    |
|-----------|-----------------|-------------------------------|--------------------------------|
| Turning   | Lathe           | Workpiece rotates, tool moves | Shafts, cylinders, knobs       |
| Drilling  | Drill Machine   | Tool rotates & feeds down     | Holes in all materials         |
| Milling   | Milling Machine | Tool rotates, workpiece moves | Flat surfaces, slots, gears    |
| Grinding  | Grinder         | Abrasive wheel rotates        | Fine finishing, hardened parts |
| Shaping   | Shaper          | Tool reciprocates             | Flat/contoured surfaces        |



6. CNC MACHINES

MODERN

DEFINITION

**CNC (Computer Numerical Control)** = Automated machining where machine tool movements are controlled by pre-programmed computer commands (G-code & M-code).

BASIC COMPONENTS

- **Input Device** – Keyboard/tape reader
- **Controller (MCU)** – Interprets program
- **Machine Tool** – Lathe/mill/drill
- **Drive System** – Servo/stepper motors
- **Feedback System** – Position sensors

ADVANTAGES

- High accuracy & repeatability
- 24/7 unmanned operation
- Complex shapes easily made
- Reduced human error
- Quick changeover

| FEATURE       | MANUAL MACHINING      | CNC MACHINING                  |
|---------------|-----------------------|--------------------------------|
| Control       | Manual by operator    | Computer-controlled            |
| Accuracy      | Operator-dependent    | Very high, consistent          |
| Speed         | Slow                  | Fast                           |
| Skilled labor | Highly skilled needed | Programmer needed              |
| Cost          | Low initial cost      | High initial cost              |
| Best for      | One-off, simple parts | Mass production, complex parts |



7. 3D PRINTING (ADDITIVE MANUFACTURING)

INDUSTRY 4.0

DEFINITION

**3D Printing / Additive Manufacturing** = Building objects layer-by-layer from digital CAD models. Opposite of subtractive machining.



COMMON METHODS

- **FDM** – Fused Deposition Modeling (plastic filament)
- **SLA** – Stereolithography (resin, UV laser)
- **SLS** – Selective Laser Sintering (powder)
- **DMLS** – Direct Metal Laser Sintering (metal)

ADVANTAGES & APPLICATIONS

-  Complex shapes without tooling
-  Customization & prototyping
-  Material saving (no waste)
-  Medical implants & prosthetics
-  Aerospace lightweight parts
-  Architecture models



8. SMART MANUFACTURING (INDUSTRY 4.0)

DEFINITION

**Smart Manufacturing** = Integration of advanced technologies (IoT, AI, robotics, big data) in manufacturing for intelligent, automated, and connected production.

## KEY CONCEPTS

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- **IoT** – Connected sensors & machines
- **Automation** – Robots replacing manual work
- **AI & ML** – Predictive maintenance
- **Digital Twin** – Virtual replica of physical system
- **Big Data** – Data-driven decisions
- **Cloud Computing** – Remote data processing

## BENEFITS

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-  Higher efficiency & productivity
-  Reduced defects & waste
-  Predictive maintenance
-  Flexible & agile production
-  Energy optimization
-  Reduced downtime



9. BOILERS

DEFINITION

**Boiler** = A closed pressure vessel in which water is heated and converted into steam at high pressure using fuel combustion. Used in power plants, industries.



| FEATURE        | FIRE TUBE BOILER    | WATER TUBE BOILER            |
|----------------|---------------------|------------------------------|
| Fluid in Tubes | Hot flue gases      | Water                        |
| Steam Pressure | Low (up to 25 bar)  | High (up to 300 bar)         |
| Efficiency     | Lower               | Higher                       |
| Size & Weight  | Bulky, heavy        | Compact, lighter             |
| Start-up Time  | Slow                | Fast                         |
| Examples       | Lancashire, Cochran | Babcock & Wilcox, Benson     |
| Use            | Small industries    | Power plants, large industry |



10. THERMODYNAMIC CYCLES

VERY IMPORTANT

FORMULAS

OTTO CYCLE – PETROL/GASOLINE ENGINE

$\eta_{Otto} = 1 - (1/r^{\gamma-1})$

r = Compression Ratio ( $V_1/V_2$ )  
γ = Ratio of specific heats ( $C_p/C_v$ ) ≈ 1.4 for air  
Processes: 1-2 Adiabatic compression, 2-3 Constant volume heat addition, 3-4 Adiabatic expansion, 4-1 Constant volume heat rejection

DIESEL CYCLE – DIESEL ENGINE

$\eta_{Diesel} = 1 - (1/r^{\gamma-1}) \times [(\rho^{\gamma}-1) / \gamma(\rho-1)]$

r = Compression Ratio  
ρ = Cut-off ratio ( $V_3/V_2$ )  
γ = 1.4 for air  
Processes: 1-2 Adiabatic compression, 2-3 Constant pressure heat addition, 3-4 Adiabatic expansion, 4-1 Constant volume heat rejection

| FEATURE           | OTTO CYCLE        | DIESEL CYCLE              | RANKINE CYCLE     |
|-------------------|-------------------|---------------------------|-------------------|
| Engine Type       | Petrol/SI Engine  | Diesel/CI Engine          | Steam Turbine     |
| Heat Addition     | Constant volume   | Constant pressure         | Constant pressure |
| Compression Ratio | 6–10              | 14–22                     | –                 |
| Efficiency        | Depends only on r | Lower than Otto at same r | 30–40%            |
| Fuel              | Petrol, CNG       | Diesel                    | Water/Steam       |
| Application       | Cars, motorcycles | Trucks, buses, ships      | Power stations    |

KEY FORMULA RELATIONSHIPS

$\eta_{Otto}$  increases as r increases | For same r:  $\eta_{Otto} > \eta_{Diesel}$  | In practice: Diesel is more efficient (higher r possible)



**Work done:**  $W = Q_{\text{supplied}} - Q_{\text{rejected}}$  | **Thermal Efficiency:**  $\eta = W_{\text{net}} / Q_{\text{supplied}}$



11. REFRIGERATION & AIR CONDITIONING

DEFINITION

**Refrigeration** = Process of maintaining a space/substance at a temperature lower than surroundings by removing heat from it. Based on vapour compression cycle.

Evaporator (Absorb Heat) → Compressor (Increase Pressure) → Condenser (Release Heat) → Expansion Valve → Evaporator

COMPONENTS & FUNCTIONS

- **Compressor** – Increases refrigerant pressure & temperature
- **Condenser** – Releases heat to surroundings; gas → liquid
- **Expansion Valve** – Reduces pressure & temperature
- **Evaporator** – Absorbs heat from space; liquid → gas
- **Refrigerant** – Working fluid (R-22, R-410A, R-134a)

KEY TERMS

- **COP** =  $Q_L / W$  (Coefficient of Performance)
- **Ton of Refrigeration** = 3.5 kW of cooling
- **AC** = Refrigeration + humidity + air circulation control
- **VCR Cycle** = Vapour Compression Refrigeration (most common)
- **VAR Cycle** = Vapour Absorption (uses heat, not electricity)



12. IC ENGINES

VERY IMPORTANT

DEFINITION

**IC Engine (Internal Combustion Engine)** = Engine where combustion of fuel occurs inside the cylinder, directly driving the piston. Heat energy → Mechanical energy.

| FEATURE             | SI ENGINE (SPARK IGNITION) | CI ENGINE (COMPRESSION IGNITION)      |
|---------------------|----------------------------|---------------------------------------|
| Full Name           | Spark Ignition Engine      | Compression Ignition Engine           |
| Fuel                | Petrol, CNG, LPG           | Diesel                                |
| Ignition            | Electric spark plug        | High compression heat (auto-ignition) |
| Compression Ratio   | 6 : 1 to 10 : 1            | 14 : 1 to 22 : 1                      |
| Thermodynamic Cycle | Otto Cycle                 | Diesel Cycle                          |
| Efficiency          | Lower (25–30%)             | Higher (35–45%)                       |
| Noise & Vibration   | Less                       | More                                  |
| Weight              | Lighter                    | Heavier                               |
| Emissions           | CO, HC, NOx                | NOx, PM, smoke                        |
| Application         | Cars, 2-wheelers, aircraft | Trucks, buses, ships, generators      |



13. 2-STROKE VS 4-STROKE ENGINES

MANDATORY

| FEATURE       | 2-STROKE ENGINE  | 4-STROKE ENGINE     |
|---------------|------------------|---------------------|
| Power Strokes | 1 per revolution | 1 per 2 revolutions |

|                    |                                |                          |
|--------------------|--------------------------------|--------------------------|
| Power Output       | More power for same size       | Less power for same size |
| Weight & Size      | Light, compact                 | Heavy, larger            |
| Valves             | No valves (ports only)         | Inlet & exhaust valves   |
| Lubrication        | Petrol + oil mixture           | Separate oil sump        |
| Fuel Efficiency    | Poor (fuel loss in scavenging) | Better                   |
| Thermal Efficiency | Lower                          | Higher                   |
| Emissions          | More smoke, pollution          | Less pollution           |
| Maintenance        | Simple, easy                   | Complex, costly          |
| Applications       | Mopeds, chainsaws, outboards   | Cars, trucks, generators |



14. ELECTRIC & HYBRID VEHICLES

FUTURE TECH

ELECTRIC VEHICLE (EV)

- Powered entirely by electric motor
- Energy stored in battery pack (Li-ion)
- Zero direct emissions
- Recharged from grid
- Ex: Tesla, Tata Nexon EV, Ola S1

HYBRID VEHICLE (HEV)

- Uses both IC engine + electric motor
- Regenerative braking recharges battery
- Better fuel efficiency than pure IC
- Lower emissions than IC only
- Ex: Toyota Prius, Honda Accord Hybrid

EV COMPONENTS

- **Battery Pack** – Energy storage (kWh)
- **Electric Motor** – Converts electrical to mechanical energy
- **Controller/Inverter** – Regulates motor power
- **Charger (OBC)** – On-board charger
- **BMS** – Battery Management System

| VEHICLE        | ENERGY SOURCE    | EMISSIONS                  |
|----------------|------------------|----------------------------|
| IC Engine      | Fossil fuel      | High CO <sub>2</sub> , NOx |
| Hybrid (HEV)   | Fuel + Battery   | Reduced                    |
| Plug-in Hybrid | Fuel + Grid      | Lower                      |
| Electric (BEV) | Grid electricity | Zero direct                |



KEY TERMS & DEFINITIONS (30+)

|                       |                                                        |                          |                                                        |
|-----------------------|--------------------------------------------------------|--------------------------|--------------------------------------------------------|
| <b>Casting</b>        | Pouring molten metal into mould to obtain shape        | <b>Forging</b>           | Shaping metal by compressive hammering force           |
| <b>Rolling</b>        | Deforming metal by passing between rollers             | <b>Extrusion</b>         | Forcing material through die to get cross-section      |
| <b>Welding</b>        | Joining metals by melting with heat/pressure           | <b>Brazing</b>           | Joining with brass filler above 450°C                  |
| <b>Soldering</b>      | Low-temp joining with tin-lead alloy below 450°C       | <b>Turning</b>           | Rotating workpiece against cutting tool on lathe       |
| <b>Drilling</b>       | Making holes using rotating drill bit                  | <b>Milling</b>           | Removing material using rotating multi-tooth cutter    |
| <b>CNC</b>            | Computer Numerical Control – automated machining       | <b>3D Printing</b>       | Additive manufacturing – layer-by-layer fabrication    |
| <b>Boiler</b>         | Closed vessel converting water to steam under pressure | <b>Otto Cycle</b>        | Thermodynamic cycle for petrol/SI engines              |
| <b>Diesel Cycle</b>   | Thermodynamic cycle for diesel/CI engines              | <b>Rankine Cycle</b>     | Steam power plant cycle (steam turbine)                |
| <b>Refrigeration</b>  | Maintaining space below ambient temperature            | <b>COP</b>               | Coefficient of Performance = $Q_L / \text{Work input}$ |
| <b>IC Engine</b>      | Engine with internal fuel combustion in cylinder       | <b>SI Engine</b>         | Spark Ignition – uses spark plug; petrol fuel          |
| <b>CI Engine</b>      | Compression Ignition – self-ignites; diesel fuel       | <b>Compression Ratio</b> | $V_{\text{max}} / V_{\text{min}}$ in engine cylinder   |
| <b>Scavenging</b>     | Removing burnt gases from cylinder (2-stroke)          | <b>Pattern</b>           | Replica of casting used to make mould cavity           |
| <b>Core</b>           | Sand insert to create hollow in casting                | <b>Hot Working</b>       | Forming above recrystallization temperature            |
| <b>Cold Working</b>   | Forming below recrystallization temperature            | <b>Work Hardening</b>    | Increase in hardness/strength due to cold working      |
| <b>Industry 4.0</b>   | 4th industrial revolution: IoT, AI, automation         | <b>Digital Twin</b>      | Virtual model mirroring a real physical system         |
| <b>Evaporator</b>     | AC component that absorbs heat to cool space           | <b>Condenser</b>         | AC component that rejects heat to surroundings         |
| <b>Hybrid Vehicle</b> | Vehicle with both IC engine and electric motor         | <b>BMS</b>               | Battery Management System – monitors EV battery        |



FORMULAS & KEY RELATIONS

EXAM CRITICAL

OTTO CYCLE EFFICIENCY

$\eta = 1 - 1 / r^{(\gamma-1)}$

r = compression ratio, γ = 1.4 (air)

DIESEL CYCLE EFFICIENCY

$\eta = 1 - (1/r^{\gamma-1}) \times [(p^{\gamma}-1)/\gamma(p-1)]$

ρ = cut-off ratio, r = compression ratio

THERMAL EFFICIENCY

$\eta_{th} = W_{net} / Q_{supplied}$   
 $= (Q_1 - Q_2) / Q_1$

COEFFICIENT OF PERFORMANCE (REFRIGERATION)

$COP_R = Q_L / W_{net}$

Q\_L = heat absorbed from cold space, W = work input

COP OF HEAT PUMP

$COP_{HP} = Q_H / W_{net} = COP_R + 1$

Q\_H = heat rejected to hot space

HEAT TRANSFER (BASIC)

$Q = m \times c_p \times \Delta T$   
m = mass (kg), c\_p = specific heat, ΔT = temp difference



IMPORTANT COMPARISON TABLES

MANUFACTURING PROCESSES COMPARISON

| PROCESS         | MATERIAL STATE | ACCURACY    | SURFACE FINISH | COST   | BEST FOR                    |
|-----------------|----------------|-------------|----------------|--------|-----------------------------|
| Sand Casting    | Liquid         | Low         | Rough          | Low    | Large complex parts         |
| Die Casting     | Liquid         | High        | Smooth         | High   | Mass production, thin walls |
| Forging         | Solid (hot)    | Medium      | Medium         | Medium | High-strength parts         |
| Machining (CNC) | Solid          | Very High   | Excellent      | High   | Precision components        |
| 3D Printing     | Powder/Liquid  | Medium-High | Good           | Medium | Prototypes, complex shapes  |

ENGINE COMPARISON

| ENGINE TYPE    | FUEL        | CYCLE   | EFFICIENCY | USE           |
|----------------|-------------|---------|------------|---------------|
| SI (Petrol)    | Petrol/CNG  | Otto    | 25–30%     | Cars, bikes   |
| CI (Diesel)    | Diesel      | Diesel  | 35–45%     | Trucks, ships |
| Electric Motor | Electricity | –       | 85–95%     | EVs, trams    |
| Steam Turbine  | Steam       | Rankine | 30–40%     | Power plants  |

JOINING METHODS COMPARISON

| METHOD  | TEMPERATURE | STRENGTH  | REVERSIBLE? | USE                           |
|---------|-------------|-----------|-------------|-------------------------------|
| Welding | Very High   | Very High | No          | Structural, heavy fabrication |
| Brazing | High        | High      | No          | Thin/dissimilar metals        |

|                  |     |            |     |                   |
|------------------|-----|------------|-----|-------------------|
| Soldering        | Low | Low        | Yes | Electronics, PCB  |
| Riveting         | RT  | Medium     | No  | Aircraft, bridges |
| Adhesive Bonding | RT  | Low–Medium | No  | Composites, glass |



## FLASHCARDS

35 CARDS

**Q:** What is casting?**A:** Process of pouring molten metal into a mould to get desired shape**Q:** What is a core in casting?**A:** Sand insert placed in mould to create hollow sections**Q:** What is forging?**A:** Plastic deformation of metal using compressive hammer/die force**Q:** What is extrusion?**A:** Forcing material through a shaped die to get uniform cross-section**Q:** What is welding?**A:** Permanent joining of metals by applying heat to melt and fuse them**Q:** What is turning?**A:** Machining where workpiece rotates and tool removes material (lathe)**Q:** What is 3D Printing?**A:** Additive manufacturing – building objects layer-by-layer from CAD model**Q:** What is a Boiler?**A:** Closed pressure vessel that converts water to steam using heat**Q:** Otto cycle is used in?**A:** Petrol/SI engines (Spark Ignition)**Q:** Otto cycle efficiency formula?**A:**  $\eta = 1 - 1/r^{(\gamma-1)}$  where  $r$  = compression ratio**Q:** SI Engine vs CI Engine ignition?**A:** SI: spark plug. CI: self-ignition due to high compression heat**Q:** 2-stroke power stroke frequency?**A:** One power stroke per revolution (every 2 strokes)**Q:** What is refrigeration?**A:** Removing heat from a space to maintain it below ambient temperature**Q:** Function of compressor in AC?**A:** Increases refrigerant pressure and temperature**Q:** What is a pattern in casting?**A:** Replica of the final product used to create mould cavity**Q:** What is die casting?**A:** High-pressure injection of molten metal into permanent metal mould**Q:** What is rolling?**A:** Deforming metal by passing between two rotating rolls**Q:** Difference between hot & cold working?**A:** Hot: above recrystallization temp. Cold: below it (work hardening occurs)**Q:** Soldering vs Brazing temperature?**A:** Soldering <450°C; Brazing >450°C with brass/bronze filler**Q:** What is CNC?**A:** Computer Numerical Control – machines controlled by pre-programmed code**Q:** What is Industry 4.0?**A:** 4th industrial revolution: IoT, AI, automation in smart manufacturing**Q:** Fire tube vs Water tube boiler?**A:** Fire tube: gases in tubes. Water tube: water in tubes (high pressure)**Q:** Diesel cycle is used in?**A:** Diesel/CI engines (Compression Ignition)**Q:** What is compression ratio?**A:** Ratio of maximum to minimum cylinder volume ( $V_{\max} / V_{\min}$ )**Q:** Which engine has higher efficiency?**A:** CI (Diesel) engine – higher compression ratio allowed**Q:** 4-stroke power stroke frequency?**A:** One power stroke per 2 revolutions (every 4 strokes)**Q:** COP of refrigerator formula?**A:**  $COP = Q_L / W$  (heat removed / work input)**Q:** Function of evaporator in AC?**A:** Absorbs heat from space to cool it; refrigerant evaporates

**Q:** What is a hybrid vehicle?

**A:** Vehicle using both IC engine and electric motor for propulsion

**Q:** What is digital twin?

**A:** Virtual replica of a physical system for simulation and monitoring

**Q:** Ton of refrigeration = ?

**A:** 3.5 kW (or 12,000 BTU/hr) of cooling capacity

**Q:** What is the Rankine cycle?

**A:** Thermodynamic cycle for steam power plants using water as working fluid

**Q:** What is FDM in 3D printing?

**A:** Fused Deposition Modeling – melting plastic filament layer by layer

**Q:** What is scavenging?

**A:** Removing exhaust gases from cylinder using fresh charge (2-stroke)

**Q:** What is MIG welding?

**A:** Metal Inert Gas welding – wire electrode + shielding gas, semi-automatic





## EXAM TIPS & STRATEGY

### ✗ COMMON MISTAKES TO AVOID

- ⚠ Confusing **SI (petrol)** and **CI (diesel)** engines – remember SI = Spark, CI = Compression
- ⚠ Writing **Diesel cycle** formula incorrectly – always include the cut-off ratio  $\rho$
- ⚠ Mixing up **Brazing** (>450°C) and **Soldering** (<450°C) temperatures
- ⚠ Forgetting **Fire tube** = gas in tubes, **Water tube** = water in tubes
- ⚠ Writing long paragraphs instead of tables – examiners prefer structured answers
- ⚠ Ignoring cycle diagrams – always draw P-V diagram for Otto/Diesel cycle

### ✓ WRITING TIPS FOR HIGH MARKS

- 💡 Use **comparison tables** wherever two or more topics asked together
- 💡 Draw simple text/schematic **diagrams** for casting, boilers, and refrigeration
- 💡 **Highlight keywords**: underline definitions first, then explain
- 💡 For engine questions: always mention **fuel, cycle, compression ratio, efficiency**
- 💡 State formula, then substitute values clearly in numerical problems
- 💡 Use bullet points for advantages/disadvantages – never paragraphs

### ⚡ HIGH-PROBABILITY EXAM TOPICS

Casting steps & types | SI vs CI engine table | 2-stroke vs 4-stroke table | Otto & Diesel cycle formulas | Refrigeration components | CNC advantages | Fire tube vs Water tube boiler | Welding types & temperatures



## QUICK REVISION SHEET – LAST 30 MINUTES



### MANUFACTURING – KEYWORDS

Casting Forging Rolling Extrusion Drawing Welding  
Soldering Brazing Riveting Turning Drilling Milling  
CNC 3D Print

### KEY DIFFERENCES TO REMEMBER

- **Hot > Recrystallization temp**; Cold < Recryst temp
- **Welding** = same metal; **Brazing** = different metals OK
- **Casting** = liquid metal; **Forging** = solid metal
- **CNC** = computer-controlled; **Manual** = operator-controlled
- **FDM** = plastic filament; **SLA** = resin+UV laser

### 🔥 THERMAL – KEYWORDS

Boiler Fire Tube Water Tube Otto Cycle Diesel Cycle  
Rankine Refrigeration COP Compressor Condenser  
SI Engine CI Engine 2-Stroke 4-Stroke EV Hybrid

### MUST-KNOW NUMBERS

- **SI engine**:  $r = 6-10$ , fuel = petrol, cycle = Otto
- **CI engine**:  $r = 14-22$ , fuel = diesel, cycle = Diesel
- **Brazing > 450°C**; Soldering < 450°C
- $\gamma = 1.4$  for air in cycle calculations
- **1 Ton ref = 3.5 kW** cooling
- **2-stroke**: 1 power/rev; **4-stroke**: 1 power/2 rev

### ✓ FINAL CHECKLIST BEFORE EXAM

☐ Know all 5 manufacturing process types with examples | ☐ Can write casting steps in order | ☐ Know SI vs CI engine differences (full table) | ☐ Know Otto & Diesel formulas + what each variable means | ☐ Know refrigeration 4 components + their functions | ☐ Know 2-stroke vs 4-stroke table | ☐ Can explain CNC advantages & 3D printing steps | ☐ Know fire tube vs water tube differences

